Arsalan Ghogari

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GitHub: https://github.com/arsalanghogari/

Education

University of Southern California: Viterbi School of Engineering, Marshall School of Business **Aug 2023 - Present**

Bachelor of Science, Computer Science/Business Administration; Minor, AI Applications

Study Abroad: University of Bristol, UK | Spring 2026

- GPA: 3.9; Presidential Scholar, Viterbi Scholar, Dean's List, W.V.T. Rusch Honors, Thematic Option Honors
- Relevant Coursework: Data Structures & Algorithms, Organizational Leadership, Communications, Business Statistics, App Development

Experience

Director of Digital Strategy, USC Undergraduate Student Government **May 2025 - Present**

- Manage and maintain USC USG's official WordPress website, improving accessibility, information architecture, and usability for over 20,000 students.
- Developed front-end features using HTML, CSS, and JavaScript, including dynamic homepage announcement systems and interactive election interfaces.
- Built and deployed Legislative Project Tracker and Senate Bill Tracker systems by integrating Google Forms, Sheets, and NinjaTables to automate transparency workflows.
- Developed an interactive Funding Tracker visualizing budget allocation across research, arts, philanthropy, and student initiatives.
- Built and deployed Ask USG, a production RAG chatbot for USC's 20,000+ undergraduate student body: Express/Node + OpenAI on Render, Postgres/pgvector retrieval, SSE streaming widget integrated with the USG WordPress site.
- Designed a deterministic content-freshness gate that stamps every knowledge chunk with its WordPress last-modified date and routes users to staff when sources predate the current year, plus a mental-health crisis pathway (keyword + moderation detection, guardrailed scripted responses) routing to the 988 Lifeline.
- Resolved complex CSV parsing and synchronization issues affecting tracker reliability and maintainability.
- Invited (and accepted) to continue serving in the role for an additional academic year in recognition of technical contributions, initiative, and impact on USC USG digital infrastructure.

Perception & Autonomy Lead / Software Engineer, USC RoboSub **Jan 2024 - Present**

- Lead development of the AUV's perception and autonomy stack, integrating computer vision, navigation, SLAM, and mission control for autonomous underwater operation.
- Collected and annotated underwater datasets to train YOLO26 and RT-DETR object detection models, improving detection performance from 0.40 to 0.70 to 0.90 mAP.
- Integrated vision with ROS-based SLAM by publishing perception outputs as ROS topics, enabling real-time localization and navigation using visual detections.
- Developed mission logic and navigation software across the autonomy stack, coordinating perception, planning, and control between ROS nodes.
- Built high-fidelity NVIDIA Isaac Sim environments modeling the AUV, cameras, and sensors with ROS integration, enabling closed-loop sim-to-real testing of the perception and navigation stack prior to in-water deployment.
- Recorded, managed, and analyzed ROS bag datasets in Foxglove to train and debug perception/navigation models, visualize sensor and detection topics, measure latency and detection metrics, and replay runs for reproducible, simulation-driven regression testing.
- Containerized ROS services with Docker and Kubernetes for deployment across NVIDIA Jetson Xavier and Raspberry Pi hardware.
- Built and maintained the team website using React, TypeScript, & Notion API, improving club visibility.

Math Peer Mentor, C.E.N.T.R.I.C. Program (USC Viterbi & Dornsife) **Aug 2025 – Dec 2025**

- Supported launch of USC Viterbi and Dornsife's mentoring initiative for first-generation and transfer students through

one-on-one mentorship and outreach.
- Collaborated with 30+ professors, advisors, and residential staff to coordinate outreach and increase student engagement.
- Managed mentee application tracking, mentor matching, scheduling, and communication workflows

Technical Projects

USC USG AI Chatbot | Node/Express, Python, OpenAI, Postgres/pgvector, LangGraph, Langfuse, GitHub Actions

- Built and shipped a production RAG chatbot for USC's Undergraduate Student Government, auto-ingesting all 97 pages/posts on the USG WordPress site into Postgres/pgvector (HNSW, idempotent upserts), refreshed weekly via GitHub Actions with zero manual steps.
- Added two-stage retrieval (20-candidate vector recall → LLM rerank), improving MRR from 0.91 to 0.98 on a 51-question golden set measured by a self-built eval harness (hit-rate, MRR, LLM-judge faithfulness) with Langfuse tracing.
- Cut time-to-first-token from ~6.2s to ~2.3s via SSE token streaming while keeping a mental-health crisis pathway whole-message and guardrail-checked ahead of every response.
- Extended into a tool-using agent (LangGraph state machine) choosing between KB retrieval and live data tools (events calendar, project tracker) at 100% tool-selection accuracy (22/22); hardened with rate limiting, CORS allowlist, input validation, Docker, and CI with an automated eval job.

AI-Powered Smart Seasoning Carousel | React, Tailwind CSS, OpenAI GPT-5.4, ESP32

- Built an AI-powered smart seasoning carousel integrating an ESP32 microcontroller with a React/Tailwind interface to streamline cooking workflows and ingredient organization.
- Integrated GPT-5.4-based recipe generation to recommend personalized meals and seasoning combinations based on available ingredients and user preferences.

Collaborative Note-Taking Application | Java, XML, Android Studio

- Developed a collaborative Android note-taking application featuring intuitive UI workflows and persistent note management functionality.
- Selected as one of the top 2 implementations out of 37 teams and invited to present the project to the class for technical and design excellence.

Personal Portfolio Website | React, TypeScript, HTML/CSS, JavaScript

- Designed and developed a responsive personal portfolio showcasing technical projects, leadership experience, and professional achievements with a focus on performance and user experience.
- Built custom interactive components and deployed the site to a custom domain (**arsalanghogari.com**), serving as a centralized hub for projects, resume, and contact information.

Study Abroad Diaries (3D Web Experience) | HTML, CSS, JavaScript

- Developed an interactive 3D photo carousel documenting experiences from studying abroad at the University of Bristol, featuring draggable animations and immersive storytelling.
- Engineered a custom JavaScript carousel with dynamic image rendering, responsive design, and accompanying narratives to create an engaging visual portfolio experience.

Skills

Programming: Python, C++, C, Java, JavaScript, TypeScript, React, HTML/CSS, MATLAB
Technologies & Frameworks: ROS, Kubernetes, Docker, OpenAI API, YOLO, WordPress, Java Servlets, Express/Node, PostgreSQL (pgvector), LangGraph, Langfuse, GitHub Actions CI/CD
Tools: Git, Google Workspace, MS Office, OnShape, AutoCAD